

FOYER ELECTRICIAN MASTER PLAN

Pre-plaster conduit + wiring + sign-off reference

Everything the electrician (and other trades) need to lay conduits, cut cavities, install back boxes, pull wires, and sign off the foyer welcome system installation BEFORE plaster goes on.

Project:	Ganesh Prasad Home, Chitradurga
Document version:	1.1 - 2026-05-22 (see Revision page next)
Source:	electrical/FOYER_MASTER_ELECTRICIAN_PLAN.md
Audience:	Electrician (primary), mason, painter, carpenter, stone cladder, false-ceiling contractor, smart-home installer, homeowner
Critical deadline:	Phase 1 marking + sign-off starts 2026-05-23. Plaster ~2026-05-30.
Sections:	18 total - design (1-10), schedules (11-14), BOM + build + sign-off (15-17), open items (18)
Pages:	approx 50 (with diagrams)
Cost estimate:	approx Rs. 1,06,230 for foyer welcome scope (excludes smart lock, dining speaker, Waveshare hardware)

CRITICAL

Plaster crew arrives ~2026-05-30. Once plaster goes on, no more changes. All conduits, back boxes, draw strings must be in place + sign-off form (Section 17) signed by homeowner BEFORE plaster starts.

The full top-down foyer floor plan with all zone markers, conduit endpoints, and device positions is in Section 2 (The Big Picture). Detailed wall elevations, cavity views, and the conduit routing map are in their respective sections.

R

Revision v1.1 (2026-05-22)

6 changes from v1.0 (2026-05-20). Read this BEFORE the detail sections
- some prose below still uses v1.0 wording; these notes override.

CRITICAL Important

If you printed v1.0 already, the 6 changes below override the detail sections. Detail sections were not all re-edited - the revision summary is the canonical v1.1 update.

The 6 changes

#	What changed	Where in doc
1	CAVITY SPEAKER -> CEILING SPEAKER. Visaton FR 8 + MDF baffle + 3 inch speaker pocket all DROPPED. 3 inch flush ceiling speaker added at foyer false-ceiling centre. PAM8403 amp stays in cavity. Speaker wire runs up Conduit #4 (16mm grey, cavity-top to false ceiling - previously pull-string-only, now 2-core speaker wire). Reason: electrician (nephew) flagged 3 inch blind pocket is hard to plaster reliably. Bonus: new monitor LS22F320GAW has no built-in speakers, so external speaker was always required.	Sec 7 Comp 4, Sec 7.6 (pocket dropped), Sec 9 (ceiling speaker added)
2	CAVITY SOCKET PANEL: 2-module -> 8-module (8M). Adds a Cat6 keystone jack for the data conduit termination (proper structured cabling). Sonoff stays inside box. Both power AND data conduits terminate in this 8M box. Socket pocket grows to ~7 x 3 x 2 inch deep.	Sec 7 Comp 5, Sec 7.6 (pocket size), Sec 14
3	FOYER SWITCH PANEL: 6-gang -> 18M VERTICAL grid. Pocket already cut on-site, fits vertically on the 1ft6inch N wall section. Extra modules reserved for porch switches + walnut shelf switch + possible foyer/living 2-way (living-hall 2-way scope deferred to electrician).	Sec 8
4	SMART-SWITCH BOX APPROACH RELAXED: depth-as-needed + 2 extra modules per Sonoff. Supersedes blanket 65mm-everywhere mandate from 2026-05-06. General rule: standard 50mm box + 2 extra modules where feasible. Exception: high-density boards (Foyer Switch Panel with 6 Sonoffs) still need 65mm + 2M-per-Sonoff per Sec 8.5.	Sec 8.5, Sec 14
5	DOORBELL: Reolink Video Doorbell PoE -> Hikvision DS-KV6113-WPE1(C). 2MP/1080p IP villa door station, 131 degree FOV, 120dB WDR, PoE, IP65. Datasheet confirms continuous RTSP + ONVIF -> works with Frigate 24/7 for the known-face auto-welcome flow (NOT press-only). Use accessory DS-KABV6113-A for theft resistance; TAMPER alarm built in. Same brand as CAM-3/4/5 -> unified NVR. Caveat: verify RTSP enabled in device config once in CAM-1 = PROVISION ONLY (confirmed). Conduit + flush-capable IP67 back box + pull string on porch W wall, NO camera installed now. Rationale: Hikvision doorbell does continuous face detection, and CAM-1 would sit only ~2-3 ft away (largely redundant). Future option: small flush anti-theft PoE pinhole if doorbell-only detection disappoints (wall stays ready).	Sec 2 hardware chain, Sec 5 Zone E, Sec 10 (all subsections), Sec 15
6		Sec 10

Other context to know

The welcome-screen MONITOR was also swapped in a separate 2026-05-26 update: LS22F350 -> Samsung LS22F320GAWXXL (VESA 100, IPS, 36.2mm deep, no built-in speakers, 1.6 kg, Rs.6,499). Sec 7 Components 1+2 already reflect this. The new thinner monitor gives ~64mm cavity rear gap (up from ~51mm), which makes the 8M socket box + RPi + amp layout comfortable.

SMART SWITCH MODULE locked to Sonoff ZBMINI R2 (Wipro NextHome + Shelly rejected per 2026-05-26 materials-checklist update).

WAVESHARE INDOOR PANEL hardware locked 2026-05-25: Raspberry Pi 5 (4GB) + WaveShare PoE HAT (F) + 10.1 inch DSI touchscreen + PN532 NFC reader + walnut recessed bezel. See [electrical/WAVESHARE_INDOOR_PANEL.md](#) for full spec.

NOTE

How to read the detail sections

Where you see 'Reolink' in the prose: read as 'Hikvision DS-KV6113-WPE1(C)' (same install spot, same Cat6 PoE wiring, different brand/model). Where you see 'cavity speaker / Visaton FR 8 / speaker pocket': those are DROPPED - read 'ceiling speaker' instead. Where you see '2-module cavity socket': read '8M with Cat6 keystone'. Where you see '6-gang switch panel': read '18M vertical'. Where you see '65mm boxes everywhere': read 'depth-as-needed + 2 modules per Sonoff'. Section 10 has a doorbell brand-swap note at the top. Section 7 Components 4 + 5 + 7.6 are fully rewritten for v1.1.



Table of Contents

Sections, audiences, and where to find each one.

Document map

#	Section	Primary audience
1	Document purpose + how to use this	Everyone
2	The big picture - what we are building	Everyone
3	Plain-English glossary	Everyone
4	Foyer geometry (corrected)	Mason + Electrician
5	The 6 installation zones (overview)	Everyone
6	East wall - DB + starter cupboard	Electrician + Carpenter
7	South wall - Screen cavity (most detail)	Electrician + Painter + Stone cladder + Carpenter
8	North wall - Switch panel + door + window	Electrician
9	Foyer ceiling - Spots + cove + drivers	Electrician + False-ceiling contractor
10	Porch + niche routes - Doorbell + cameras + Waveshare	Electrician
11	Complete conduit schedule (master table)	Electrician
12	Wire schedule (every wire, gauge, colour)	Electrician
13	MCB plan (2 MCBs for foyer)	Electrician
14	Switch board + socket schedule	Electrician
15	Bill of materials (line-item shopping list)	Homeowner + Electrician
16	Phase-by-phase build order	All trades
17	Pre-plaster sign-off checklist	Homeowner + Electrician
18	Flagged open items (decide later)	Homeowner

How to use this document

Each section is self-contained. Jump to the wall or zone you are working on. Section 17 is the FINAL gate before plaster - tick off every item before the plaster crew arrives. Section 11 is the master conduit list - the electrician ticks off each conduit as it gets laid.

01

Document Purpose

What this document is, who reads what, and the critical timeline.

What this document is

The single source of truth for the foyer welcome system installation. It contains everything needed to lay the right conduits in the right places BEFORE plastering, cut the right cavities, pull the right wires, and install the right switches, sockets, cameras, doorbell, and screens.

Who reads what

Trade	Sections
Electrician (in this region, also cuts cavity + DB recess + chases + lays conduits + pulls wires + installs switches)	4-14, 16, 17
Mason (plastering after conduits in, stone wall building if applicable)	4, 16, 17
Painter (cavity interior matte black, side walls, ceiling)	7, 9
Carpenter (VESA mount, cupboard facade, Waveshare frame, walnut shelf)	6, 7, 8, 10
Stone cladder (S feature wall + cavity reveal wrap)	7, 16
False-ceiling contractor (gypsum, cove pocket, inspection hatch)	9
Smart-home installer (commissioning, camera config)	2, 5-10
Homeowner (decisions + sign-off)	All; especially 3, 4, 17, 18

NOTE Regional practice

In Chitradurga construction, the ELECTRICIAN does cavity cutting, DB recess cutting, switch box cutting, and conduit chasing using core drills, wall grinders, and chisels. The mason's role is primarily plastering after the electrical work and any stone/brickwork. Trade assignments in this document reflect this practice.

Critical timeline

Date	Event	Status
2026-05-20	Document v1.0 published, design locked	Done
2026-05-21 to 22	On-site walkthrough with electrician + carpenter	Done
2026-05-23	PHASE 1 starts: marking + sign-off	Imminent
2026-05-23 to 24	Phase 2: wall chasing + cavity cutting	
2026-05-25 to 26	Phase 3: conduit pulling	
2026-05-27 to 28	Phase 4: wire pulling + DB install	
2026-05-29 to 31	Phase 5: first plaster (rough coat)	

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~2026-06-15	Welcome system live + tested	TARGET

Date

CRITICAL

Once plaster goes on, you cannot change anything
All conduits, back boxes, draw strings must be in place and the Section 17 sign-off form must be signed by the homeowner BEFORE the plaster crew arrives. There is no second chance.

02

The Big Picture

What the welcome system does and how the pieces connect.

Visitor approaches the main door. The face camera sees them. Two seconds later the screen in the foyer feature wall lights up with their name and a chime plays from a hidden cavity speaker. They feel expected. They feel welcomed. That is the whole idea.

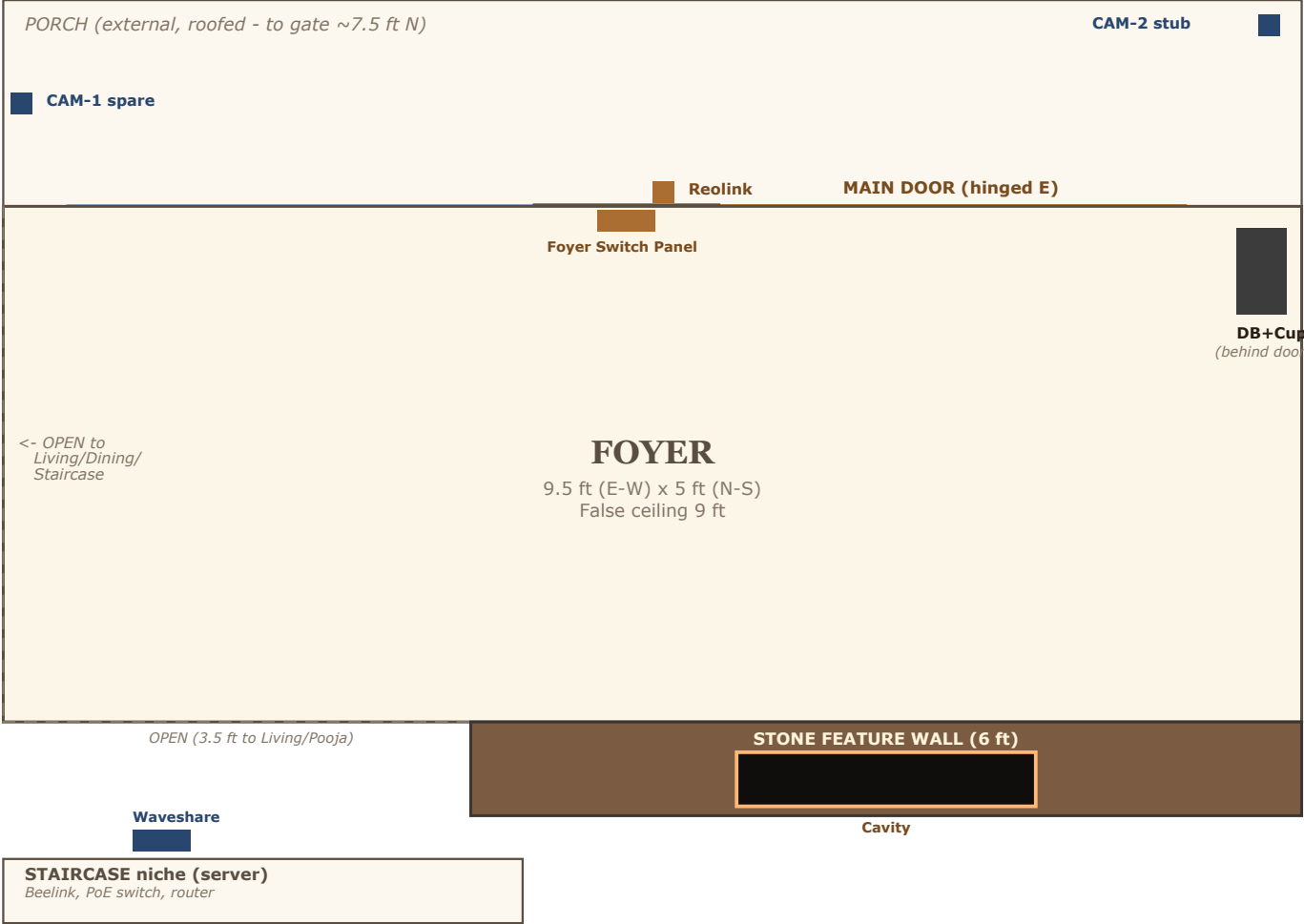
Three-step welcome flow

- SEES: Reolink Video Doorbell on the N wall 1'6" section captures the visitor's face as they approach the main door.
- RECOGNISES: Beelink EQ12 in the staircase niche runs Frigate + CompreFace - matches face within ~2 seconds.
- GREETES: Samsung 21.5" monitor in the cavity lights up; halo LED glows warm amber around the screen; speaker plays a chime.

Hardware chain (data flow)

#	Component	Where it lives	What it does
1	Reolink Video Doorbell PoE	Outside N wall, 1'6" section, 1450 mm FFL	Captures visitor's face + rings bell
2	Cat6 (PoE)	From doorbell -> staircase niche	Data + power
3	PoE switch (TP-Link TL-SG1210P)	Staircase niche	Powers cameras + Reolink + future cams
4	Beelink EQ12 (Frigate + CompreFace + HA)	Staircase niche	Identifies the face in 1-2 sec
5	Home Assistant automation	Same server	Decides what to display
6	Cat6 to cavity (PoE)	Niche -> floor -> cavity	Pushes welcome payload + power for RPi
7	Raspberry Pi Zero 2W	Inside cavity, behind monitor	Drives the monitor + speaker
8	Samsung LS22F350 monitor	Recessed in feature wall, centre 1450 mm FFL	Shows the welcome
9	Pi Camera Module 3 (CAM-0)	Top bezel of monitor	Secondary face check at close range
10	Halo LED strip	Inside cavity at 30 mm depth step, all 4 inner walls	Warm amber glow around monitor
11	Visaton FR 8 speaker + PAM8403 amp	Inside cavity, behind monitor	Chime + welcome voice from screen
12	Foyer Switch Panel (6-gang)	N wall, 1'6" section, 1200 mm FFL	Manual control + scene override

Visual: welcome system layout (top-down foyer floor plan)



03

Plain-English Glossary

Every technical term used in this document, defined.

Electrical terms

Term	Plain-English meaning
DB (Distribution Board)	The panel of breakers in the wall. Every wire starts here. Ours is on the East wall of the foyer.
MCB (Miniature Circuit)	A trip switch inside the DB. One per circuit. Trips on overload or short.
RCBO / RCCB	Like an MCB but also trips on leakage to earth. Mandatory for electronics + wet
Circuit	One wire run from a single MCB feeding a group of devices.
FFL (Finished Floor Level)	Top of the installed floor tile. All heights are measured from FFL, NOT raw slab.
Conduit	A plastic pipe inside the wall. Wires get pulled through it. Red = 230V power, Grey = low voltage.
Chase	The groove cut into the wall where the conduit sits.

Switch / socket terms

Term	Plain-English meaning
Gang	One switch position on a plate. A 4-gang plate has 4 separate switches.
Switch Board / Switch Panel	The wall plate + back box that holds the switches. We call ours the 'Foyer Switch
Modular plate	The removable face plate that snaps onto a back box.
Back box	The metal box recessed into the wall behind the plate. 65 mm DEEP for smart
5A socket	Standard 3-pin Indian outlet, 230V, for low-current things (chargers, TVs, monitors).
16A socket	Bigger 3-pin outlet for heavy appliances (geyser, AC, microwave).
Smart switch	A switch that takes commands over WiFi/Zigbee in addition to manual press.
Sonoff	WiFi smart relay that fits behind the switch plate as a hidden unit.

Light + LED terms

Term	Plain-English meaning
LED strip	A thin sticky strip with tiny LED dots. Comes in 24V or 12V. Sold in 5m rolls.
Driver (LED driver)	A small electronic brick that converts 230V AC to 24V DC for LED strips. Like a phone charger.
Halo LED	The warm amber strip stuck around the cavity inner edges. Glows around the monitor.
Cove LED	The 2700K strip hidden in the cove pocket around the false ceiling perimeter.
Spotlight (GU10)	A focused beam. GU10 bulbs run on 230V mains directly - NO driver needed.
CCT	Colour temperature. 2200K = candle warm amber, 2700K = warm white.

Camera / network terms

Term	Plain-English meaning
PoE	Power over Ethernet - sending power AND data over the same Cat6 cable.
Cat6	Category 6 Ethernet cable. Carries gigabit data + PoE power.
LSZH	Low Smoke Zero Halogen - jacket material that doesn't produce toxic gas in fire.
UV-resistant	Cable jacket that doesn't crack in sunlight. For outdoor / exposed cables.
RTSP	Real-Time Streaming Protocol - the video stream format used by IP cameras.
NVR	Network Video Recorder - records + processes camera streams. Ours = Beelink EQ12 with Frigate.
Home Assistant (HA)	The smart-home brain software. Reads sensors, decides actions, controls devices.

Cavity / door terms

Term	Plain-English meaning
Cavity	The rectangular hole cut into the S feature wall for the monitor.
VESA bracket	Metal plate (75x75 mm pattern) that screws into the back of the monitor + onto the
Baffle (MDF)	Small wooden plate that holds the speaker driver.
Vascal	The wall return between the door jamb and the East exterior wall corner.
Niche	The server cupboard built into the staircase wall.
Hinge side	Where the door hinges are - ours is EAST.
Latch side	Where the door handle is - ours is WEST.

04

Foyer Geometry

Exact dimensions, walls, openings - the foundation everything else builds on.

Foyer dimensions

Dimension	Imperial	Metric	Notes
N-S depth (door to feature wall)	5 ft (60")	1524 mm	Fixed - S feature wall is built
E-W at S edge (stone feature wall)	6 ft (72")	1828 mm	The stone wall portion
E-W at S edge (opening portion)	3.5 ft	1066 mm	Open to Living/Pooja
E-W at N edge (door wall)	9.5 ft (114")	2895 mm	Total N exterior wall length
GF ceiling height (slab)	11 ft	3353 mm	Raw slab
Foyer false ceiling height	9 ft	2743 mm	Drop of 2 ft from slab
S feature wall thickness	9"	228 mm	Structural - NOT a 4" partition
N exterior wall thickness	9"	228 mm	Exterior brick/block masonry
E exterior wall thickness	9"	228 mm	Exterior

Key reference heights (from FFL)

Reference	Inches	mm	Why
Cavity centre (screen eye level)	57-1/8"	1450	Standing eye level
Cavity bottom edge	50-1/2"	1280	Ergonomic recess
Cavity top edge	63-7/8"	1620	Auto-derived
Walnut shelf top surface	35-3/8"	900	Below cavity
Reolink doorbell centre	57"	1450	Face level for adult
Foyer Switch Panel centre	47-1/4"	1200	Standard switch height
DB bottom edge	59"	1500	MCBs at chest height
False ceiling height	108"	2743	9 ft drop from 11 ft slab

Orientation convention

When this document says LEFT or RIGHT for the cavity or any wall, it means: you are standing in the foyer, looking AT the wall in question. LEFT/RIGHT is from your perspective.

For the South feature wall (looking SOUTH at the cavity from foyer):

LEFT = East side (the DB direction)

RIGHT = West side (the staircase niche direction)

For the North wall (looking NORTH at the door from inside foyer):

LEFT = West side (where the corner window is)

05

The 6 Installation Zones

High-level map of all the work zones in the foyer welcome scope.

Zone	Wall / area	Detail section	Key items
A	East wall (interior)	6	DB + starter cupboard
B	South feature wall	7	Screen cavity + walnut shelf
C	North wall	8	Switch panel + door + window
D	Foyer ceiling	9	Spots + cove + halo + drivers
E	Porch (external)	10	Reolink + CAM-1 spare + CAM-2 stub
F	Staircase indoor	10	Waveshare + server niche

Zones A through D are inside the foyer. Zone E is on the porch (outside the foyer but under the FF balcony slab overhang). Zone F is in the staircase area which is indoor but outside the foyer. All 5 external/indoor-non-foyer conduits originate from the staircase niche.

Conduit count per zone

Zone	Conduits
A (East wall / DB)	2 going OUT of DB + 5 incoming from water automation system
B (South cavity)	3 entering cavity (Power, Data, Halo 24V) + 1 provision exiting (ceiling speaker future)
C (North switch panel)	1 incoming from DB + 4-6 outgoing to loads (Spots, Cove+Halo, Shelf, Porch Ceiling, Porch Wall, Spare)
D (Ceiling)	Branch conduits from switch panel + driver-to-strip
E (Porch external)	3 long-runs from niche (Reolink, CAM-1 spare, CAM-2 stub)
F (Staircase indoor)	1 short-run (Waveshare)

06

East Wall - DB + Starter Cupboard

Zone A. DB recessed in wall; starter cupboard surface-mounted alongside or below.

Equipment on the East wall

Item	Dimensions (W x H x D)	Mount	Centre / bottom height
DB recess (Schneider Acti9 48-way)	400 x 600 x 100 mm	Flush recessed	Bottom edge 1500 mm
Starter cupboard (P1 + P2 + Sonoff DUALR3 + terminals)	600 x 400 x 250 mm	SURFACE-MOUNT TE D	TBD on-site

The geometric constraint

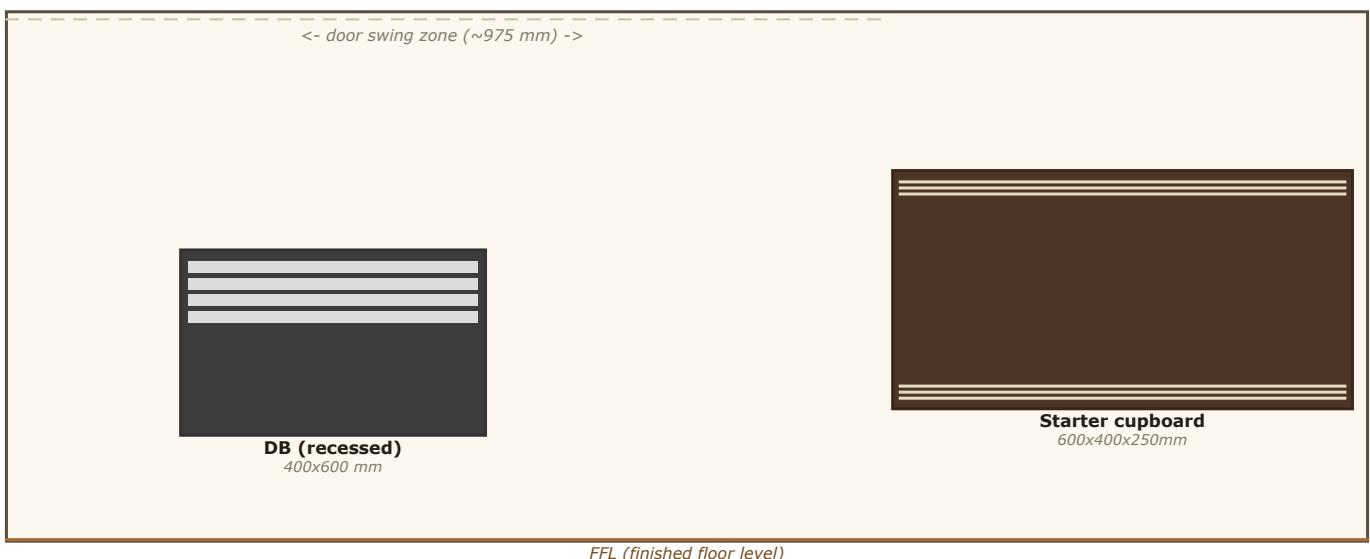
East wall is 5 ft (1524 mm) long N-S. Main door (3.2 ft = 975 mm wide) swings against East wall when open. Door panel covers the first 975 mm from the N corner. That leaves only 549 mm of East wall South of the door panel arc.

DB (400 mm wide) + Cupboard (600 mm wide) side-by-side need 1000 mm. They DON'T fit in 549 mm. Solution: stack them vertically OR DB sits behind door panel + cupboard south of swing (with small overhang).

NOTE **Electrician decides on-site**

Final layout (Option 1 = side-by-side with cupboard slight overhang OR Option 2 = stacked vertically inside door swing zone) is chosen on-site after actual measurements. Homeowner signs off on the chosen layout.

East wall elevation (visual)

EAST WALL ELEVATION (interior view)

Conduits originating from the DB (only 2 go out!)

Only TWO conduits actually start at the DB to feed the foyer. Within each, multiple wires run together (L+N+E). Loads are then fed downstream from the Foyer Switch Panel - NOT directly from the DB.

#	Conduit ID	Size	To	Wires inside
1	C-DB-Foyer-Switch	25 mm	Foyer Switch Panel (N wall, 1200 FFL)	1.5 sqmm L + N + E
2	C-DB-Cavity-Power	25 mm	Cavity bottom-LEFT corner of back wall	2.5 sqmm L + N + E

RED

Conduits ENTERING the cupboard (water automation)

Cross-reference: water-automation-conduits.md

#	Conduit ID	Size	From	Carries
1	C-Sintex-2	16 mm	Sintex tank float (terrace)	2-core 1.5 sqmm 220V
2	C-Sump-2	16 mm	Sump JB (east external wall)	2-core 1.5 sqmm 220V
3	C-DB-Backup	20 mm	Staircase server niche	Empty pull string
4	C-Motor-P1	25 mm	Outside (borewell)	3-core 4 sqmm armoured
5	C-Motor-P2	25 mm	Outside (P2 pump)	3-core 2.5 sqmm

Pre-plaster sign-off for Zone A

- ☐ Final layout (Option 1 or 2) selected on-site by electrician + carpenter
- ☐ Homeowner sign-off photographed
- ☐ DB recess cut to 400 x 600 x 100 mm, flush, ready for DB shell
- ☐ Cupboard mounting position marked with ventilation considerations
- ☐ All 5 outgoing DB conduits routed and labelled
- ☐ All 5 incoming water automation conduits terminated at cupboard back panel
- ☐ Draw strings (nylon twine) inserted in every conduit
- ☐ All conduit ends capped with masking tape
- ☐ Earth wire (4 sqmm) routed from earth pit to DB
- ☐ Mounting bracket / rawl plugs for DB shell ready

07

South Wall - Screen Cavity

Zone B. The most complex zone. Stone reveal wraps INTO the cavity for a premium framed look.

Cavity overview

Think of the cavity as a small medicine cabinet built into the wall. The cavity has 5 surfaces inside: TOP inner wall, BOTTOM inner wall, LEFT inner wall, RIGHT inner wall, and BACK wall. Plus the OPEN front (where the monitor hangs). Stone slips are applied to all 4 INNER walls for the FRONT 30 mm of depth only. The back wall stays matte black across its full surface.

Cavity cut dimensions

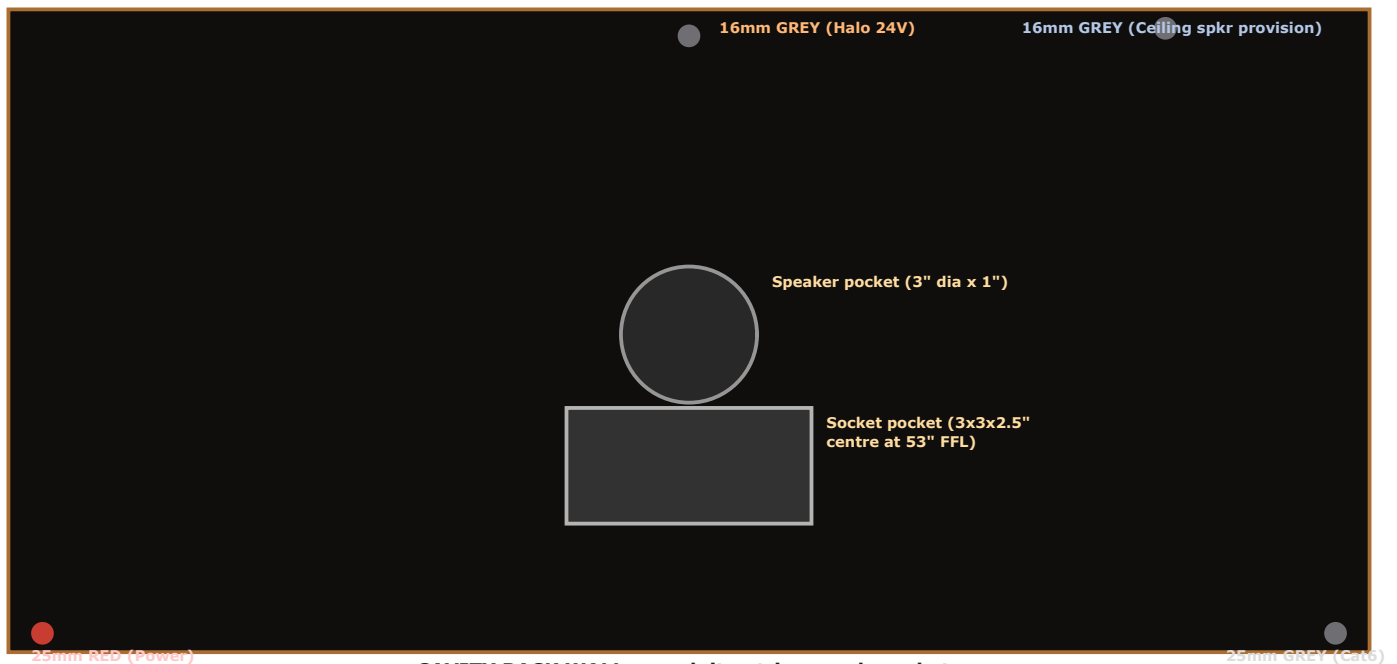
Dimension	Imperial	Metric	Notes
Cavity cut WIDTH (raw masonry)	22-3/4"	580 mm	Bigger than final, to fit stone slips
Cavity cut HEIGHT (raw masonry)	14-7/8"	380 mm	Same reason
Cavity DEPTH (into 9" wall)	4"	100 mm	Leaves 5" of masonry behind
Inner usable opening AFTER stone slips	21-1/4" x 13-1/2"	540 x 340 mm	What the monitor sees
Stone reveal depth (front portion)	1-1/4"	30 mm	First 30mm of inner walls clad in
Stone slip thickness	3/4"	20 mm	Per inner wall

Cavity front view (with monitor + halo)



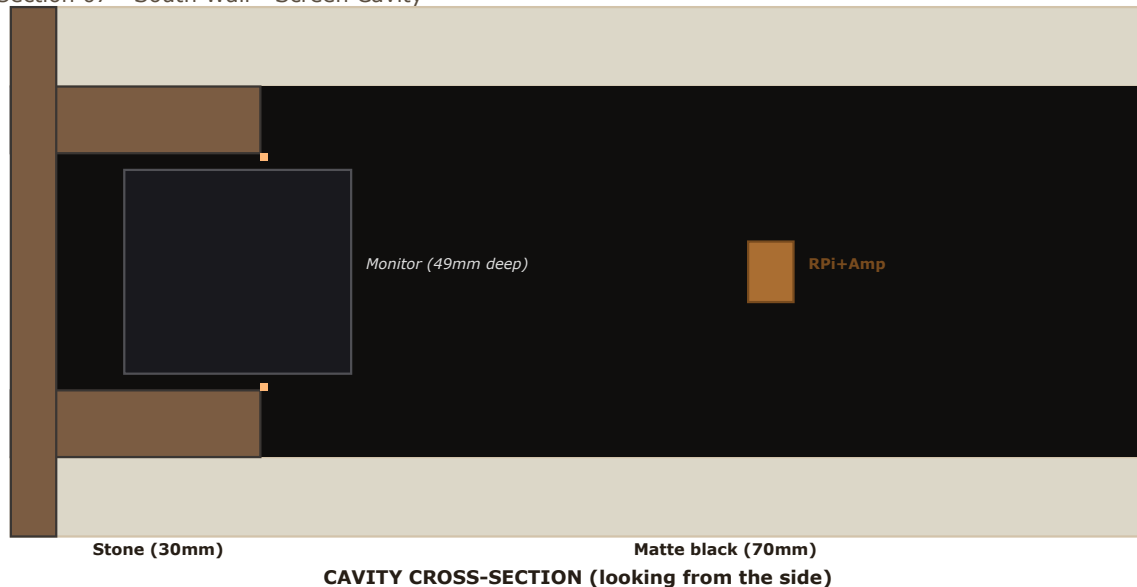
CAVITY (front view) - monitor with halo glow
 Inner opening 540x340mm (after stone reveal) - Centre 1450mm FFL

Cavity back wall - sub-pockets + conduit entries



CAVITY BACK WALL - conduit entries + sub-pockets

Stone reveal cross-section



Sub-pockets cut INTO the cavity walls

#	Pocket	Wall	Dimensions	Depth INTO ma	Position
1	Socket pocket	Cavity back wall	3" x 3"	2-1/2"	Centred horiz; centre at 53" FFL
2	Speaker pocket	Cavity back wall	3" diameter	1"	Centred horiz; centre at 57-1/8" FFL

The 4 conduits at the cavity (3 IN through back wall + 1 OUT through top)

#	Conduit	Size	Colour	Entry / exit point	From / To
1	POWER	25 mm	RED	Bottom-LEFT of back wall (1.5" up + 2" in)	From DB (East wall)
2	DATA	25 mm	GREY	Bottom-RIGHT of back wall (1.5" up + 2" in)	From staircase niche (via floor)
3	HALO 24V	16 mm	GREY	Top-CENTRE of back wall (3/4" down)	From halo driver (above false ceiling)
4	Ceiling speaker provision	16 mm	GREY	TOP wall, going UP	To future ceiling speaker (pull string only)

Components inside the cavity

#	Component	Position	Spec
1	Monitor (Samsung LS22F350)	Hangs on VESA bracket; front face flush with stone face	21.5", 491 x 291 x 49 mm, VESA 75
2	VESA bracket	Centre of back wall, 1450 mm FFL	Slim flat-mount, direct rawl plugs OR small 8"x4" ply
3	RPi Zero 2W + PoE splitter	Stuck to back wall via 3M tape	Powered by PoE + data via same Cat6
4	Visaton FR 8 speaker + MDF baffle (PRIMARY)	Back wall, behind monitor centre; magnet in 1" pocket	80 mm dia, fires forward through monitor back
4	Ceiling speaker (FUTURE OPTION)	Above false ceiling (if added later via Conduit #4)	16mm grey conduit + pull string ready now
5	Cavity Socket Panel (2 sockets)	Back wall, centre at 53" FFL, in 3"x3"x2.5" pocket	Monitor + Spare; Sonoff Mini hidden inside box

6	Bezel CAM-0 (Pi Camera Module 3)	Top-centre of monitor bezel	CSI ribbon to RPi (no conduit needed)
7	Halo LED strip (24V 2200K)	Inner walls at 30 mm depth step; ALL 4 sides	Hidden behind stone reveal lip; ~2 m total

Painter scope inside cavity

Back wall (all)	Matte black acrylic emulsion, 2 coats - BEFORE stone cladding
Inner walls DEEPER than 30 mm	Matte black, 2 coats
Inner walls FRONT 30 mm	DO NOT paint - stone slips go here
VESA backing ply (if used)	Matte black, 1 coat before install
MDF speaker baffle	Matte black, 1 coat before install

Stone cladder scope

Main wall cladding	Rustic ledgerstone (tobacco + sandstone + charcoal mix), 6 ft x 11 ft S feature wall face
Cavity reveal slips	20 mm thick stone slips on FRONT 30 mm of all 4 inner walls. Stops cleanly at 30 mm depth line.
DO NOT	Apply stone deeper than 30 mm; cover the conduit stubs; cover socket / speaker pockets.

Pre-plaster sign-off for Zone B

- ☐ Cavity cut to 22-3/4" x 14-7/8" x 4" (580 x 380 x 100 mm)
- ☐ Cavity centred on 6 ft feature wall - 24-5/8" from each side edge ON ACTUAL MEASURED WIDTH
- ☐ Bottom edge at 50-1/2" (1280 mm) FFL
- ☐ Socket pocket cut: 3" x 3" x 2-1/2" deep, centred horiz, centre at 53" FFL
- ☐ Speaker pocket cut: 3" diameter x 1" deep, centred horiz, centre at 57-1/8" FFL
- ☐ Conduit #1 (Power) hole: 1" diameter, bottom-LEFT, 1.5" up + 2" in
- ☐ Conduit #2 (Data) hole: 1" diameter, bottom-RIGHT, 1.5" up + 2" in
- ☐ Conduit #3 (Halo) hole: 5/8" diameter, top-CENTRE, 3/4" down
- ☐ Conduit #4 (Ceiling speaker provision) exit: 5/8" diameter through TOP inner wall going up
- ☐ All 4 conduit stubs protrude 1-1.5" into cavity, capped, draw strings, labelled
- ☐ Stone cladder briefed on the 30 mm reveal specification
- ☐ Painter briefed: do NOT paint front 30 mm of inner walls
- ☐ Cavity dimensions photographed
- ☐ Walnut shelf bracket positions chalked at 900 mm FFL on S wall

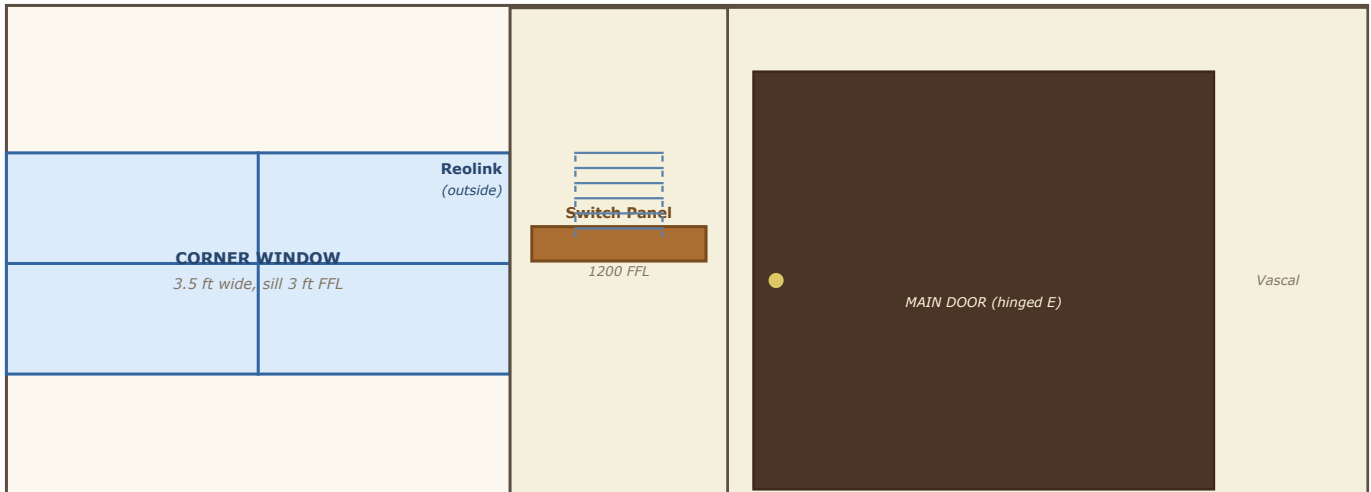
08

North Wall - Switch Panel + Door + Window

Zone C. The entry wall.

N wall layout

NORTH WALL ELEVATION (looking from inside foyer)



Wall breakdown (W to E)

Section	Width	Contains
Corner window	3'6" (1066 mm)	Sill 3 ft FFL, height 5 ft; wraps NW corner extending N
Wall section	1'6" (457 mm)	Foyer Switch Panel (inside face) + Reolink doorbell (outside face)
Door zone	4'6" (1372 mm)	Door 3.2 ft + vascal ~1'4" on East side
TOTAL	9'6" (2895 mm)	

Foyer Switch Panel - position + specs (REVISED 2026-05-23)

Property	Value
Wall section	1'6" section between window and door
Centre height (FFL)	1200 mm (47-1/4") - standard switch height
Back box (CUT ON SITE)	18M VERTICAL GI MS, ~290 x 135 x 65 mm OR 225 x 195 x 65 mm
Orientation	VERTICAL - longer side runs top to bottom
Box depth	65 mm mandatory - confirm with tape (electrician said >=2 in)
Wall capacity	1'6" = 457 mm; 18M vertical is 135 mm wide - fits with 322 mm spare
Plate	Schneider Unica 18M White (matching house-wide brand)

Slack inside box	12M slack accommodates 6 hidden Sonoffs + neutral bus (+2M per Sonoff rule)
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Gang assignments

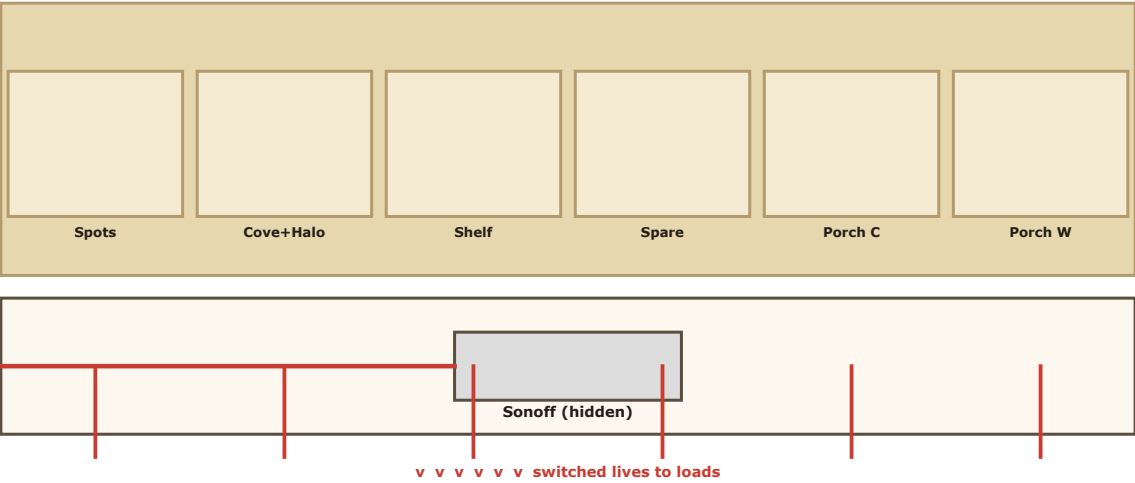
Gang	Label	Controls
1 (left, near window)	Foyer Spots	2x GU10 ceiling spotlights
2	Cove + Halo	Cove LED 2700K + Halo LED 2200K (linked)
3	Shelf	Walnut shelf under-LED strip
4	Foyer Spare	Future foyer addition (capped)
5	Porch Ceiling	Porch ceiling light(s)
6 (right, near door)	Porch Wall	Porch wall lights (future provision)
7 (if 7-gang)	Spare 7	Future use
8 (if 8-gang)	Spare 8	Future use

Conduits in / out of Switch Panel

Conduit	Size	Direction	Route
C-DB-Foyer-Switch	25 mm RED	INCOMING from DB	Up E wall -> false ceiling -> across to N wall -> drops to box
C-SW-Spots	25 mm RED	OUT (Gang 1 -> spots)	Up N wall -> false ceiling -> to 2 spot ceiling boxes
C-SW-Cove-Halo	25 mm RED	OUT (Gang 2 ->)	Up N wall -> false ceiling -> splits to cove + halo
C-SW-Shelf	16 mm RED	OUT (Gang 3 -> shelf)	Down N wall -> floor route -> up S wall -> to shelf
C-SW-Spare	16 mm RED	OUT (Gang 4 ->)	Up N wall -> false ceiling -> capped, pull string
C-SW-Porch-Ceiling	16 mm RED	OUT (Gang 5 -> porch ceil)	Up N wall -> through 9" wall -> porch ceiling
C-SW-Porch-Wall	16 mm RED	OUT (Gang 6 -> porch wall)	Up N wall -> through 9" wall -> outside near door (capped)

Switch panel wiring (visual)

FOYER SWITCH PANEL (6-gang) - wiring detail



Walnut shelf (on S wall, electrically wired from Gang 3)

Property	Value
Material	Solid walnut OR veneered MDF
Dimensions	6 ft x 10" deep x 1-1/4" thick
Top surface height (FFL)	900 mm (35-3/8")
Under-LED strip	24V, 6 W/m, 2700K, 1.8 m
LED driver location	Option 1: above false ceiling (default). Option 2: under shelf (alternative).

Pre-plaster sign-off for Zone C

- ☐ N wall total length verified (~9'6" / 2895 mm)
- ☐ Door opening cut to 3'2" wide x 7 ft tall
- ☐ Window opening cut to 3'6" wide x 5 ft tall (corner wrap)
- ☐ Vascal wall section confirmed (~1'4" East of door)
- ☐ 1'6" wall section between door and window confirmed
- ☐ Foyer Switch Panel back box: 18M VERTICAL GI MS, ~290 x 135 x 65 mm (or 225 x 195 x 65), centred horiz, centre 1200 mm FFL - confirm 65 mm depth with tape
- ☐ C-DB-Foyer-Switch conduit terminated at back box
- ☐ All outgoing conduits from switch panel routed
- ☐ Draw strings in all conduits
- ☐ All conduit ends capped
- ☐ Reolink doorbell back box installed on OUTSIDE face of 1'6" section, 1450 mm FFL
- ☐ Homeowner sign-off photographed

09

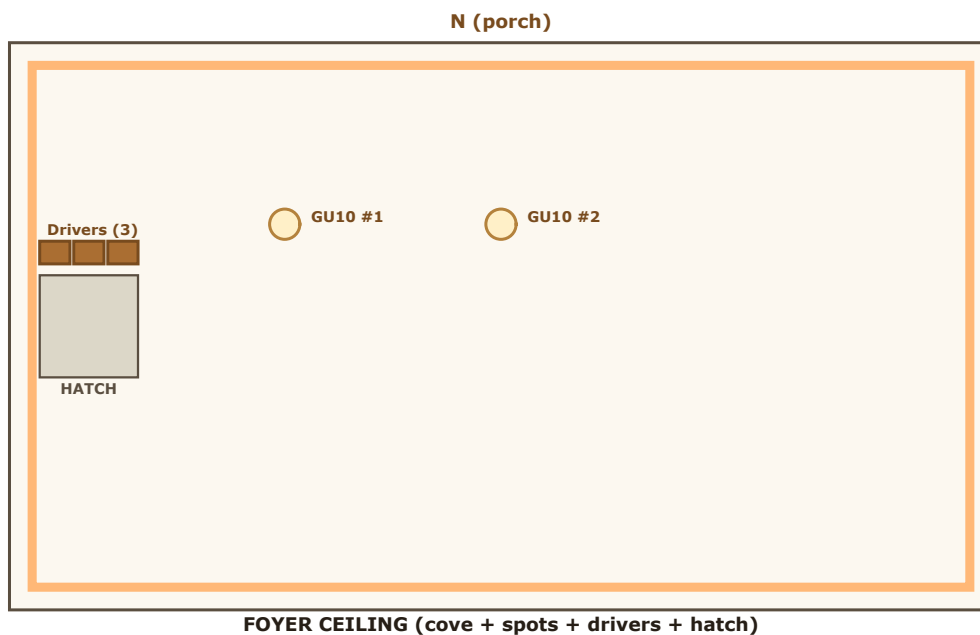
Foyer Ceiling - Spots, Cove, Drivers

Zone D. False ceiling at 9 ft. Cove pocket around perimeter. Inspection hatch above W edge.

False ceiling specs

Property	Value
Slab height	11 ft (3353 mm) FFL
False ceiling height	9 ft (2743 mm) FFL
Drop	2 ft (610 mm)
Material	12.5 mm gypsum board on GI grid
Cove pocket	100 mm wide x 75 mm deep around full perimeter
Inspection hatch	300 x 300 mm above W edge of foyer ceiling

Reflected ceiling plan



Ceiling spotlights (GU10) - NO driver needed!

NOTE **GU10 LED bulbs do NOT need a separate driver**

Each GU10 bulb has the driver electronics built into the bulb itself. They run on 230V AC mains directly. Unlike the cove / halo / shelf strips (which DO need 24V drivers), the GU10 spots wire directly from the Foyer Switch Panel Gang 1.

Property	Value
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Fitting	Recessed GU10 gimbal (adjustable angle)
Bulb	7W LED, 2700K, CRI 90+, 30 degree beam
Spot #1 position	300 mm from W wall area, 609 mm from N edge of foyer
Spot #2 position	300 mm from W wall area, 1218 mm from N edge of foyer
Aim	Graze the S feature wall at ~1700 mm FFL
Driver	NONE - built into the GU10 bulb
Wire	Direct 230V from Gang 1 (switched live + N + E)

Cove LED strip + driver

Property	Value
Strip	24V DC, 9.6 W/m, 2700K, CRI 90+, IP20
Length	~10 m around full perimeter
Position	Inside cove pocket, pointing UPWARD
Driver	Mean Well APV-100-24 (100W, 24V)
Driver location	Above false ceiling, beside inspection hatch
Control	Switched by Gang 2 (linked with halo)

Halo + Shelf LED drivers

Driver	Location	Powers
Halo driver (Mean Well APV-12-24)	Above false ceiling beside cove driver	Cavity halo strip
Shelf driver (Mean Well APV-12-24)	Option 1: above false ceiling. Option 2: under shelf.	Walnut shelf strip

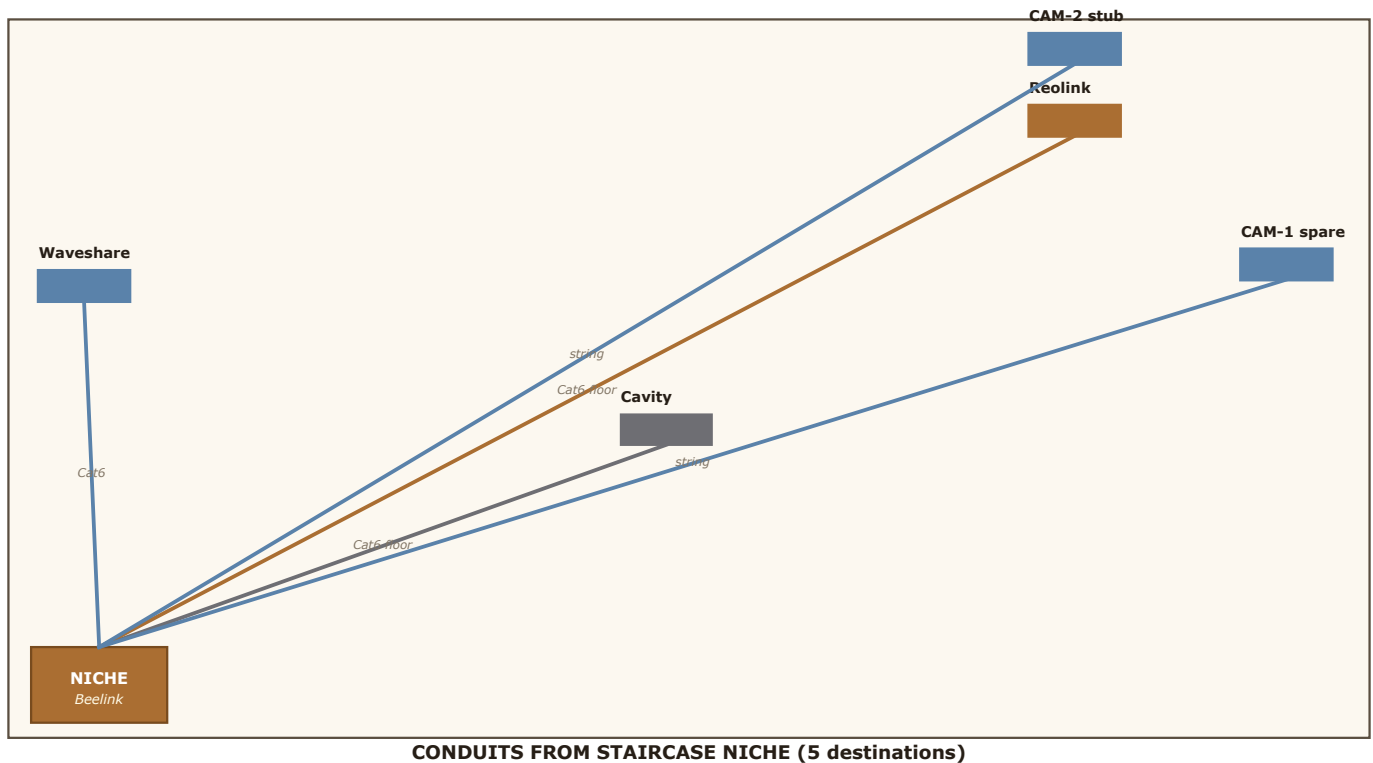
Pre-plaster sign-off for Zone D

- ☐ False ceiling drop confirmed at 2 ft (from 11 ft slab to 9 ft FFL)
- ☐ Cove pocket dimensions: 100 mm wide x 75 mm deep, around full perimeter
- ☐ Inspection hatch position marked (300 x 300 mm, W edge)
- ☐ 2x GU10 ceiling box positions chalked on raw slab
- ☐ All conduits arriving at false ceiling terminated and labelled
- ☐ Driver shelf location marked (above hatch)
- ☐ Cove strip routing path confirmed (continuous loop)
- ☐ Homeowner sign-off photographed

10 Porch + Niche Routes

Zones E + F. All external + indoor-non-foyer conduits originate at the staircase niche.

Conduit routing overview



5 conduits from the niche

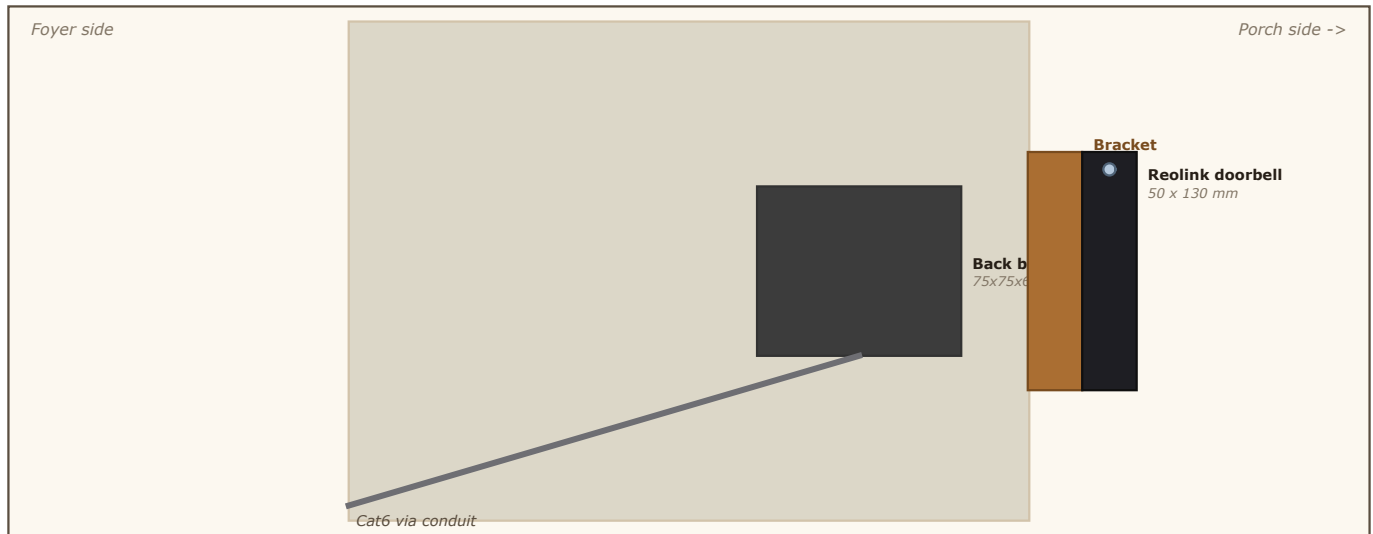
#	Conduit ID	Size	Primary route	To
1	C-Niche-Cavity-Data	25 mm	FLOOR	Cavity bottom-right (Cat6 for RPi)
2	C-Niche-Reolink	25 mm	FLOOR	Reolink doorbell on N wall outside (1450 FFL)
3	C-Niche-CAM1	25 mm	FLOOR	Porch W wall capped (1650 FFL) - pull string only
4	C-Niche-CAM2	25 mm	CEILING (only	Porch soffit NE corner (~2700 FFL) - pull string
5	C-Niche-Waveshare	25 mm	Short vertical	Staircase S wall (1500 FFL) - Cat6 pulled

GREY

CRITICAL Floor-route conduits MUST be laid before tiles

Conduits #1, #2, #3 all share the same floor chase from niche going East. The electrician digs ONE channel in the floor screed and lays 3 conduits side-by-side. But this MUST be done BEFORE the tile contractor arrives. Once tiles are down, no more floor conduits.

Reolink doorbell mount detail



REOLINK MOUNT - back box hidden behind bracket hidden behind doorbell

Reolink doorbell back box specs

Property	Value
Doorbell unit dimensions	2" wide x 5-1/8" tall (50 x 130 mm)
Bracket footprint	~50 x 130 mm (matches doorbell)
Back box (default - electrician's)	3" x 3" x 2-1/2" GI MS modular (75 x 75 x 65 mm)
Why smaller than 4x4	Must be smaller than bracket - so box stays HIDDEN behind bracket after install
Position centre (FFL)	1450 mm (57") - face level for adults
Distance from each side of 1'6" section	9" (228 mm) - centred
Mounting	Flush recessed from OUTSIDE face; rim flush with finished plaster
Conduit entry	Bottom or side, 25mm grey
Weather protection	Apply silicone sealant around conduit entry (modular box not natively IP-rated; porch is roofed)

Reolink installation procedure summary

Phase 1 (BEFORE plaster): mark position, cut 3" x 3" cavity in outside wall, cut conduit chase through 9" wall, install back box, fit cable, leave 300 mm tail, cap.

Phase 2 (AFTER plaster): strip Cat6, crimp RJ45, test with cable tester, plug into Reolink, mount bracket over back box, clip doorbell on. Configure in Reolink app + HA.

NOTE

What to buy + when

Generic 3x3x2.5 GI MS back box (Rs.100, BEFORE plaster). Silicone sealant tube (Rs.100). Outdoor LSZH UV Cat6 (~Rs.280). Order Reolink Video Doorbell PoE NOW so you can verify the bracket fits over the back box before plaster covers everything. NO separate conduit for chime - HA plays chime through cavity speaker / Waveshare / any networked device. Optional Reolink wireless chime device (Rs.1500) plugs into any 5A socket.

Pre-plaster sign-off for Zones E + F

- ☐ All 5 conduits routed from niche to their endpoints
- ☐ Floor segment of conduits #1, #2, #3 laid BEFORE tile contractor arrives - CRITICAL
- ☐ Reolink back box installed at 1450 mm FFL on outside of 1'6" N wall section
- ☐ CAM-1 spare back box installed on porch W wall, 1650 mm FFL, capped
- ☐ CAM-2 stub back box installed on porch soffit NE corner, ~2700 mm FFL, capped
- ☐ Waveshare back box installed on staircase S wall, 1500 mm FFL
- ☐ Cat6 cables pulled (Reolink, foyer cavity, Waveshare) - outdoor LSZH UV Cat6 for outdoor runs
- ☐ Pull strings inserted in unused conduits (CAM-1 spare, CAM-2 stub)
- ☐ All conduit ends capped
- ☐ Homeowner sign-off photographed for all 5 conduit routes + endpoints

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Master Conduit Schedule

Every conduit in one place. Print this and tick off as you lay each one.

Conduit colour code

Colour	Used for	Sizes
RED PVC	230V power	25 mm (heavy) and 16 mm (light loads)
GREY PVC	Low voltage - Cat6, sensor wires, LED 24V secondary	25 mm (Cat6) and 16 mm (sensor, LED)
BLUE PVC	Mains incomer + sockets/AC/heavy circuits	25 mm

If coloured PVC unavailable in Chitradurga, wrap each conduit end with coloured electrical tape AND label with permanent marker BEFORE plaster covers them.

Master conduit table

#	Conduit ID	Size	From	To	Length	Cable inside
1	C-Main-Incomer	25mm	Utility meter	DB	~5m	Mains per BESCOM
2	C-Earth	16mm	Earth pit	DB earth bus	~3m	4 sqmm Green/Yellow
3	C-DB-Foyer-Switch	25mm	DB	Foyer Switch Panel	~6m	3x 1.5 sqmm L+N+E
4	C-DB-Cavity-Power	25mm	DB	Cavity bottom-LEFT	~3m	3x 2.5 sqmm L+N+E
5	C-Niche-Cavity-Data	25mm	Niche	Cavity bottom-RIGHT (via	~12m	1x outdoor LSZH UV
6	C-Halo-24V	16mm GREY	Halo driver (above	Cavity TOP-CENTRE	~2m	2-core 0.75 sqmm 24V DC
7	C-Cavity-Ceil-Sprk (provision)	16mm GREY	Cavity TOP wall	Above false ceiling	~2m	Pull string only
8	C-SW-Spots	25mm	Switch Panel	2x spotlight boxes	~5m	3x 1.5 sqmm
9	C-SW-Cove-Halo	25mm	Switch Panel	Cove + halo drivers	~5m	3x 1.5 sqmm
1	C-SW-Shelf	16mm	Switch Panel	Shelf area	~5m	3x 1.5 sqmm
1	C-SW-Spare	16mm	Switch Panel	Above ceiling (capped)	~3m	Pull string
1	C-SW-Porch-Ceiling	16mm	Switch Panel	Porch ceiling	~4m	3x 1.5 sqmm
1	C-SW-Porch-Wall	16mm	Switch Panel	Outside near door (capped)	~4m	Pull string
1	C-Cove-24V	16mm	Cove driver	Cove strip start	~3m	2-core 1.5 sqmm 24V
1 5	C-Shelf-24V	16mm GREY	Shelf driver	Under-shelf strip	~3m (Opt 1) or 0.3m (Opt	2-core 0.75 sqmm 24V DC
1 6	C-Niche-Reolink	25mm GREY	Niche	Reolink doorbell outside (via floor)	~7m	1x outdoor LSZH UV Cat6
1 7	C-Niche-CAM1	25mm GREY	Niche	Porch W wall CAM-1 spare (via floor)	~7m	Pull string only

1	C-Niche-CAM2	25mm	Niche	Porch soffit CAM-2 stub (ceiling)	~7m	Pull string only
1	C-Niche-Waveshare	25mm	Niche	Staircase S wall Waveshare	~2m	1x outdoor LSZH UV
9		GREY				Cat6

Conduit purchasing totals

Conduit type	Total length	Buy a...
25 mm RED PVC	~17 m	30 m roll
16 mm RED PVC	~26 m	40 m roll
25 mm GREY PVC (LV-25)	~31 m	40 m roll
16 mm GREY PVC (LV-16)	~10 m	15 m roll
25 mm BLUE PVC	~5 m	10 m roll

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Wire Schedule

What goes inside each conduit. Use this to plan wire purchasing.

Wire colour code (Indian standard - mandatory)

Colour	Function	Where
RED	LIVE (phase, 230V)	Every live wire from MCB outward
BLUE	Switched LIVE leg	Between switch and the load
BLACK	NEUTRAL	Every switch board, every socket
GREEN with YELLOW stripe	EARTH	Every switch, socket, metal back box, metal fitting

Power wires (230V)

For conduit	Wire	Colours	Total length
C-DB-Foyer-Switch	1.5 sqmm	Red + Black + Green/Yellow	~18 m (3-core)
C-DB-Cavity-Power	2.5 sqmm	Red + Black + Green/Yellow	~9 m (3-core)
C-SW-Spots	1.5 sqmm	Blue + Black + Green/Yellow	~15 m
C-SW-Cove-Halo	1.5 sqmm	Blue + Black + Green/Yellow	~15 m
C-SW-Shelf	1.5 sqmm	Blue + Black + Green/Yellow	~15 m
C-SW-Porch-Ceiling	1.5 sqmm	Blue + Black + Green/Yellow	~12 m
C-Earth	4 sqmm	Green/Yellow only	~3 m

Low-voltage 24V DC wires

For conduit	Wire	Length
C-Halo-24V	2-core 0.75 sqmm 24V DC	~2 m
C-Cove-24V	2-core 1.5 sqmm 24V DC	~13 m (incl strip route)
C-Shelf-24V	2-core 0.75 sqmm 24V DC	~3 m (Opt 1) or 0.3 m (Opt 2)

Network cables (Cat6)

For conduit	Wire	Length
C-Niche-Cavity-Data	Outdoor LSZH UV Cat6	~12 m
C-Niche-Reolink	Outdoor LSZH UV Cat6	~8 m
C-Niche-Waveshare	Outdoor LSZH UV Cat6	~3 m
Subtotal outdoor Cat6		~23 m - BUY A 30 m ROLL

Spare conduits (pull string only)

Conduit	Pull string length
C-SW-Spare-4	3 m
C-SW-Porch-Wall	4 m
C-Cavity-Ceiling-Speaker	2 m
C-Niche-CAM1	7 m
C-Niche-CAM2	7 m
Subtotal	~23 m (buy a 100 m roll - cheap)

Cavity-internal wires (no conduit - all short)

Wire	Purpose	Length
3.5 mm audio cable	RPi USB sound card -> PAM8403 amp input	30 cm
USB-A to micro-USB	RPi USB port -> PAM8403 power	30 cm
2-core 1 sqmm speaker wire	Amp output -> Visaton FR 8 speaker	30 cm
CSI ribbon cable	RPi CSI port -> Pi Camera Module 3 (bezel cam)	25 cm (comes w/ cam)
Mini-HDMI to HDMI	RPi HDMI output -> monitor HDMI input	30 cm
USB-C cable	PoE splitter 5V -> RPi power input	20 cm
RJ45 patch cable	PoE splitter -> USB OTG Ethernet adapter	20 cm
Monitor power cord	Cavity Socket A -> Samsung monitor adapter	50 cm

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MCB Plan

2 dedicated MCBs for the foyer welcome system.

NOTE Electrician's verification

MCB selection below is the homeowner's plan based on load calculations. The on-site electrician should verify these against their experience + IS 8623. The 2-MCB structure (Lights + Cavity) should remain regardless of brand or exact rating choice.

2 MCBs for the foyer

MCB	Type	Rating	Protects
B-Foyer-Lights	6A MCB (B-curve)	6 A	All foyer + porch lighting (spots, cove, halo, shelf, porch lights)
B-Foyer-Cavity	6A MCB + 30mA RCBO	6 A + leakage	Cavity sockets (Monitor + Spare) - electronics + leakage protection

B-Foyer-Lights detail

MCB	6 A B-curve (Schneider / ABB / Legrand)
RCBO needed?	NO (no wet / electronic loads on lighting)
Loads	2x GU10 spots (14W) + cove ~96W + halo ~10W + shelf ~11W + porch ceiling ~9W = ~140W total
Current draw	140W / 230V = ~0.6 A (well below 6A rating)
Wire from MCB	1.5 sqmm
Via conduit	C-DB-Foyer-Switch -> Foyer Switch Panel -> 4-7 branches

B-Foyer-Cavity detail

MCB	6 A B-curve
RCBO	YES - 30 mA RCBO (mandatory for electronics)
Loads	Monitor ~25W + RPi via PoE (0W from this) + halo via shared driver + speaker amp ~1W + spare = ~36W typical
Current draw	~0.16 A typical
Wire from MCB	2.5 sqmm
Via conduit	C-DB-Cavity-Power -> directly to cavity socket box

DB layout



Earth + safety

Item	Spec
Main earth pit	Separate copper plate pit (not shared with motor pit)
Earth wire pit -> DB	4 sqmm Green/Yellow via C-Earth conduit
Earth bonding	All metal back boxes + plates + socket earth terminals -> DB earth busbar
Continuity test	< 1 ohm between any metal body and earth bus (verify with multimeter BEFORE plaster)

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Switch Board + Socket Schedule

Every back box, location, height. Total 10 boxes.

Inside foyer

#	Box	Location	Centre FFL	Size	Contents
1	Foyer Switch Panel	N wall, 1'6" section, centred	1200 mm	18M VERT 290x135x65 (cut on site)	6 rockers + 6 Sonoff ZBMINI R2 hidden

Inside the cavity

#	Box	Location	Centre FFL	Size	Contents
2	Cavity Socket Panel	Cavity back wall, in 3x3 pocket	53" (1346 mm)	75x75x65 modular	2x 5A sockets (Monitor + Spare) + Sonoff hidden

On porch (outside, weather-exposed but roofed)

#	Box	Location	Centre FFL	Type	Contents
3	Reolink doorbell back box	Outside face of 1'6" N wall, centred	1450 mm	3x3x2.5 GI MS modular	Cat6 termination behind doorbell
4	CAM-1 spare back box	Porch W wall	1650 mm	4x4 IP67 outdoor	Empty - pull string only
5	CAM-2 stub back box	Porch soffit NE corner	~2700 mm	4x4 IP67 outdoor	Empty - pull string only
6	Porch ceiling light	Centre of porch ceiling	~10 ft	B-type ceiling rose	LED fitting TBD
7	Porch wall light box	Outside near door (future)	1900-2100 mm	Standard wall box	Empty - pull string

Indoor staircase

#	Box	Location	Centre FFL	Size	Contents
8	Waveshare back box	Staircase S wall	1500 mm	TBD (custom)	Cat6 termination for future Waveshare

Ceiling boxes (inside false ceiling)

#	Box	Position	Type	Contents
9	Spotlight #1	609 mm from N edge, 300 mm from W area	B-type, 60 mm depth	GU10 gimbal fitting

#	Box
1	Spotlight #2
0	

Total back boxes: 10

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Bill of Materials

Line-item shopping list. Prices approximate (Chitradurga / online, mid-2026).

A. Conduits + accessories

Item	Quantity	Cost
25 mm RED PVC conduit (ISI)	30 m roll	Rs. 400
16 mm RED PVC conduit	40 m roll	Rs. 350
25 mm GREY PVC conduit (LV-25)	40 m roll	Rs. 450
16 mm GREY PVC conduit (LV-16)	15 m roll	Rs. 150
25 mm BLUE PVC conduit (mains)	10 m	Rs. 150
Conduit bends 25mm + 16mm	40 pieces	Rs. 250
Conduit couplers	30 pieces	Rs. 140
Conduit clamps / saddles	50 pieces	Rs. 150
Nylon pull string	100 m roll	Rs. 150
Masking tape (for capping)	2 rolls	Rs. 80
Subtotal A		~Rs. 2,270

B. Wires (foyer-scope allocation; coils shared with whole house)

Item	Quantity	Brand	Cost
1.5 sqmm Red (90m coil)	1 coil	Polycab / Havells	Rs. 1,200
1.5 sqmm Blue (switched live)	1 coil	Polycab / Havells	Rs. 1,200
1.5 sqmm Black	1 coil	Polycab / Havells	Rs. 1,200
1.5 sqmm Green/Yellow	1 coil	Polycab / Havells	Rs. 1,200
2.5 sqmm Red + Black + Green/Yellow	shared	Polycab	Rs. 6,600
4 sqmm Green/Yellow (main earth)	5 m	Polycab	Rs. 400
2-core 0.75 sqmm 24V LED	10 m	Local LED supply	Rs. 300
2-core 1.5 sqmm 24V (cove)	15 m	Local LED supply	Rs. 450
Outdoor LSZH UV Cat6	30 m	D-Link / Commscope	Rs. 1,200
Indoor Cat6 patch	2 m	Any	Rs. 50
Subtotal B			~Rs. 13,800

C. Back boxes + plates

Item	Quantity	Cost
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6-gang GI MS modular back box (default)	1	Rs. 250
2-module modular back box (cavity socket)	1	Rs. 120
3x3 GI MS modular box (Reolink default)	1	Rs. 100
4x4 IP67 outdoor box (CAM-1 + CAM-2)	2	Rs. 600
B-type ceiling rose box	3	Rs. 150
Standard wall box (porch wall future)	1	Rs. 80
Waveshare back box (custom)	1	Rs. 200
Schneider Unica 18M White plate + frame	1	Rs. 650
2-module plate	1	Rs. 150
5A sockets	2	Rs. 200
Cover plates / blanks	4	Rs. 100
Subtotal C		~Rs. 2,550 - 5,050

D. MCBs + DB hardware

Item	Quantity	Cost
6A B-curve MCB	2	Rs. 500
30 mA RCBO	1	Rs. 2,000
100 mA S-type RCCB	1 (shared)	Rs. 2,500
DB enclosure (Schneider Acti9 48-way)	1	Already planned
Copper earth busbar	1	Already planned
MCB labels	1 set	Rs. 150
Subtotal D		~Rs. 5,150

E. Smart hardware

Item	Quantity	Cost
Sonoff ZBMINI R2 x 6 (hidden behind 6 rockers in 18M)	6	Rs. 4,200
Sonoff Mini R2 (cavity socket)	1	Rs. 800
RJ45 connectors (Cat6)	20 pieces	Rs. 100
Cat6 cable tester	1	Rs. 500
Crimping tool (RJ45)	1	Rs. 400
Subtotal E		~Rs. 3,300

F. Welcome screen + cavity components

Item	Quantity	Cost
Samsung LS22F350 monitor (21.5" IPS, VESA 75)	1	Rs. 9,000
VESA 75 slim wall bracket	1	Rs. 500
Raspberry Pi Zero 2W	1	Rs. 2,500

Raspberry Pi Camera Module 3 (CAM-0)	1	Rs. 2,500
PoE splitter (5V USB-C output)	1	Rs. 1,000
USB OTG Ethernet adapter	1	Rs. 400
USB sound card (dongle)	1	Rs. 400
PAM8403 amplifier module	1	Rs. 150
Mini-HDMI to HDMI cable (30 cm)	1	Rs. 200
3.5 mm audio cable (30 cm)	1	Rs. 100
3M VHB tape + Velcro + silicone + cable ties	various	Rs. 600
Subtotal F		~Rs. 17,350

G. Audio

Item	Quantity	Cost
Visaton FR 8 speaker driver (80 mm)	1	Rs. 1,800
MDF baffle (3.5" x 3.5" x 1/4")	1	Rs. 50
2-core speaker wire	1 m	Rs. 50
Subtotal G		~Rs. 1,900

H. Lighting

Item	Quantity	Cost
GU10 LED bulb (7W, 2700K, dimmable)	2	Rs. 1,000
GU10 gimbal fitting	2	Rs. 2,000
24V LED strip 2200K (5m roll - halo)	1 roll	Rs. 1,000
24V LED strip 2700K (5m roll - shelf)	1 roll	Rs. 800
24V LED strip 2700K (10m roll - cove)	1 roll	Rs. 1,500
Mean Well APV-12-24 driver (halo + shelf)	2	Rs. 1,200
Mean Well APV-100-24 driver (cove)	1	Rs. 1,500
LED strip connectors + clips	1 set	Rs. 200
Subtotal H		~Rs. 9,200

I. Cameras + doorbell

Item	Quantity	Cost
Reolink Video Doorbell PoE (TC541)	1	Rs. 8,000
Reolink wireless chime (OPTIONAL hardware backup)	0 or 1	Rs. 1,500
(Future) CAM-1 PoE camera	0 - provision only	Future Rs. 15,000
(Future) CAM-2 overview camera	0 - provision only	Future Rs. 12,000
Subtotal I (now)		~Rs. 8,000

J. False ceiling + carpentry

Item	Quantity	Cost
Inspection hatch (300x300 mm)	1	Rs. 500
Walnut floating shelf (6 ft x 10" x 1.25")	1	Rs. 4,500
Hidden steel L-brackets for shelf	4	Rs. 400
Cupboard plywood facade	1	Rs. 3,000
Cabinet hinges + lock	per design	Rs. 500
Gypsum board + GI grid	per spec	Included in FC contractor
Subtotal J		~Rs. 8,900

K. Stone cladding

Item	Quantity	Cost
Rustic ledgerstone slips	~6 sqm	Rs. 15,000
Extra stone slips for cavity reveal	~0.5 sqm	Rs. 1,250
MS polymer adhesive (high-bond)	5 kg	Rs. 1,500
Mechanical anchor clips	per spec	Rs. 500
Subtotal K		~Rs. 18,250

L. Labour (foyer welcome scope only)

Trade	Hours	Cost
Electrician (cuts cavity + DB + conduits + wiring + termination)	40 hr	Rs. 8,000
Painter (cavity matte black + side walls)	8 hr	Rs. 2,000
Stone cladder	12 hr	Rs. 4,800
Carpenter (shelf + cupboard facade + baffle)	6 hr	Rs. 1,800
False-ceiling contractor	per spec	Rs. 6,000
Smart-home installer (Phase 11 commissioning)	8 hr	Rs. 4,000
Subtotal L		~Rs. 26,600

Grand total estimate

TIP Foyer welcome system - all-inclusive

Conduits Rs.2,270 + Wires Rs.2,760 (allocation) + Boxes Rs.2,550 + MCBs Rs.5,150 + Smart Rs.3,300 + Screen Rs.17,350 + Audio Rs.1,900 + Lighting Rs.9,200 + Doorbell Rs.8,000 + FC + Carpentry Rs.8,900 + Stone Rs.18,250 + Labour Rs.26,600 = approx Rs. 1,06,230. Excludes smart lock, dining hall speaker, Waveshare hardware (flagged in Section 18 for separate sessions).

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Phase-by-Phase Build Order

11 phases. Each has owner, duration, deliverable, and sign-off.

Timeline

Date	Phase	Owner	Status
2026-05-23	Phase 1: Marking + sign-off	Electrician + Carpenter + Homeowner	Imminen +
2026-05-23 to	Phase 2: Wall chasing + cavity cutting	Electrician	
2026-05-25 to	Phase 3: Conduit pulling	Electrician	
2026-05-27 to	Phase 4: Wire pulling + DB install	Electrician	
2026-05-29 to	Phase 5: First plaster rough coat	Mason	
2026-06-01	Phase 6: Cavity matte black	Painter	
2026-06-01	Phase 7: VESA backing + carpenter prep	Carpenter	
2026-06-02 to	Phase 8: Stone cladding incl. reveal wrap	Stone cladder	
2026-06-06 to	Phase 9: False ceiling + cove + drivers	FC contractor + Electrician	
2026-06-10 to	Phase 10: Final paint	Painter	
2026-06-13 to	Phase 11: Smart-home commissioning	Electrician + Smart-home installer	
2026-06-15	Welcome system live + tested	TARGET	

Phase 1 - Marking (Day 1)

Owner: Electrician + Carpenter + Homeowner. Duration: 1 day. On bare brick / block walls, chalk-mark every: cavity outline, DB recess, switch box positions, socket box positions, conduit chase lines, spotlight ceiling positions on slab, Reolink doorbell position on outside face, CAM-1 spare + CAM-2 stub + Waveshare back box positions. Take photos of every mark. Homeowner signs off BEFORE any cutting.

Phase 2 - Chasing + cavity cutting (Day 2-3)

Owner: Electrician. Tools: angle grinder, chisel + hammer, core drill, wall chaser. Cut main cavity (22-3/4 x 14-7/8 x 4"); socket pocket (3x3x2.5"); speaker pocket (3" dia x 1"); DB recess (15-3/4 x 23-5/8 x 4"); 6-gang switch panel pocket (335x75x65); all other back box pockets; Reolink doorbell pocket on porch wall; all conduit chases vertical + horizontal.

Phase 3 - Conduit pulling (Day 4-5)

Lay all 19 main conduits per Section 11 schedule. Use coloured PVC where available, otherwise label with tape + permanent marker. Insert pull strings in every conduit. Cap all ends with masking tape. FLOOR-ROUTE CONDUITS (#1, #2, #3) MUST be laid in floor screed BEFORE tile contractor arrives.

Phase 4 - Wire pulling + DB install (Day 6-7)

Pull wires through conduits per Section 12. Every smart-switch box gets L+N+E. Mount Schneider Acti9 48-way DB. Connect MCBs + RCBOs per Section 13. Leave 300 mm tail of wires inside every switch box. Test earth continuity ($< 1 \text{ ohm}$). Pull outdoor Cat6 through niche-to-cavity, niche-to-Reolink, niche-to-Waveshare.

Phases 5-11 (after plaster)

Phase	What happens
5 - Plaster	Apply rough coat. Conduits + back boxes flush with finished plaster line. Cure 48 hours.
6 - Matte black	Paint cavity interior matte black (back wall + deeper 70mm of inner walls). NOT front 30mm.
7 - VESA backing	Drill VESA pattern into masonry OR install small ply backing. Cut MDF speaker baffle.
8 - Stone	Clad 6 ft x 11 ft S feature wall + WRAP stone slips into cavity reveal (front 30mm, 4 sides).
9 - False ceiling	Drop ceiling to 9 ft. Build cove pocket. Install inspection hatch. Mount spotlights. Install drivers.
10 - Final paint	N wall + side walls in warm ivory. Ceiling + cove inside in soft snow. Mask stone wall + boxes.
11 - Commissioning	Install switches + sockets + plates + Sonoffs. Mount monitor + RPi + CAM-0 + speaker. Configure Frigate + CompreFace + HA. Train 3 family faces. Test welcome flow.

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Pre-Plaster Sign-Off Checklist

The MASTER CHECKLIST. Tick each item on-site BEFORE plaster crew arrives. Print + sign.

CRITICAL Read this first

Once plaster goes on the walls, NOTHING about the welcome system can be changed. This checklist is the FINAL gate. Print these pages, walk the site item by item, tick each box, and sign at the end. The plaster crew does NOT start until the homeowner has signed.

General (before any zone sign-off)

- ☐ FFL chalk line drawn on every wall (so heights can be verified from real floor level)
- ☐ All conduit labels readable (tape + marker on every conduit end)
- ☐ All pull strings inserted in spare / empty conduits
- ☐ All conduit ends capped with masking tape
- ☐ All back boxes set with rims slightly recessed (so plaster goes around rim, not on top)

Zone A - East wall (DB + starter cupboard)

- ☐ DB recess cut: 400 x 600 x 100 mm, flush, bottom 1500 mm FFL
- ☐ Starter cupboard mount position marked (Option 1 or 2 chosen on-site)
- ☐ Homeowner signed off on layout choice
- ☐ All 5 outgoing DB conduits routed and labelled
- ☐ All 5 incoming water-auto conduits terminated at cupboard back panel
- ☐ Draw strings in every conduit; ends capped
- ☐ DB shell mounting bracket / rawl plugs ready
- ☐ Earth wire (4 sqmm) routed from earth pit to DB

Zone B - South feature wall (Screen Cavity)

- ☐ Cavity cut to 22-3/4" x 14-7/8" x 4" (580 x 380 x 100 mm)
- ☐ Cavity centred on 6 ft feature wall (ON ACTUAL MEASURED WIDTH)
- ☐ Bottom edge at 50-1/2" (1280 mm) FFL
- ☐ Socket pocket cut: 3" x 3" x 2-1/2" deep, centred horiz, centre at 53" FFL
- ☐ Speaker pocket cut: 3" diameter x 1" deep, centred horiz, centre at 57-1/8" FFL
- ☐ Conduit #1 (Power) hole: 1" diameter, bottom-LEFT, 1.5" up + 2" in
- ☐ Conduit #2 (Data) hole: 1" diameter, bottom-RIGHT, 1.5" up + 2" in
- ☐ Conduit #3 (Halo) hole: 5/8" diameter, top-CENTRE, 3/4" down
- ☐ Conduit #4 (Ceiling speaker provision) exit through TOP inner wall going up
- ☐ All 4 cavity conduit stubs protrude 1-1.5" into cavity, capped
- ☐ Stone cladder briefed on 30 mm reveal depth
- ☐ Painter briefed: do NOT paint front 30 mm of inner walls
- ☐ Cavity dimensions photographed

Zone C - North wall (Switch Panel + door + window)

- ☐ N wall total length verified (~9'6" / 2895 mm)
- ☐ Door opening cut to 3'2" wide x 7 ft tall
- ☐ Window opening cut to 3'6" wide x 5 ft tall (corner wrap)
- ☐ Vascal wall section confirmed (~1'4" East of door)
- ☐ 1'6" wall section between door and window confirmed
- ☐ Foyer Switch Panel back box: 6-gang (or 7/8 if it fits), 335 x 75 x 65 mm, centred horiz, centre 1200 mm FFL
- ☐ C-DB-Foyer-Switch conduit terminated at back box
- ☐ All outgoing conduits from switch panel routed (4 mandatory + 2 porch + spare slots)
- ☐ Draw strings in all conduits
- ☐ Reolink doorbell back box installed on OUTSIDE face of 1'6" section, 1450 mm FFL
- ☐ Cat6 pulled through C-Niche-Reolink, 300 mm tail in box, capped

Zone D - Foyer ceiling

- ☐ False ceiling drop confirmed at 2 ft (from 11 ft to 9 ft FFL)
- ☐ Cove pocket dimensions: 100 mm wide x 75 mm deep, around full perimeter
- ☐ Inspection hatch position marked (300 x 300 mm, W edge)
- ☐ 2x GU10 ceiling box positions chalked on raw slab (609 + 1218 mm from N edge)
- ☐ All conduits arriving at false ceiling terminated and labelled
- ☐ Driver shelf location marked above inspection hatch

Zone E - Porch (external)

- ☐ All 3 porch conduits routed from niche (Reolink, CAM-1, CAM-2)
- ☐ FLOOR SEGMENT of all 3 floor-route conduits laid BEFORE tile contractor arrives [CRITICAL]
- ☐ Reolink back box (outside 1'6" N wall, 1450 mm FFL)
- ☐ CAM-1 spare back box (porch W wall, 1650 mm FFL) - capped
- ☐ CAM-2 stub back box (porch soffit NE corner, ~2700 mm FFL) - capped
- ☐ Pull strings in unused conduits (CAM-1, CAM-2)
- ☐ Cat6 pulled in C-Niche-Reolink (300 mm tail in back box)
- ☐ All conduit ends capped

Zone F - Staircase indoor

- ☐ Waveshare back box installed at 1500 mm FFL on staircase S wall
- ☐ C-Niche-Waveshare conduit terminated at box
- ☐ Cat6 pulled through, 300 mm tail in box, capped

Photographs to take BEFORE plaster (minimum 25 photos)

Store all photos in 'Inconstruction images/foyer-preplaster/' folder for permanent reference. At minimum: East wall DB area (3 angles); cavity front view + close-up of each conduit hole + close-up of sub-pockets; N wall elevation + switch panel; Reolink position outside; CAM-1 + CAM-2 + Waveshare back boxes; ALL 3 floor-route conduits before tile; false ceiling conduit ends.

HOMEOWNER SIGN-OFF FORM

Date of sign-off: _____

I have verified the following:

- ☐ All zones above are checked off
- ☐ All photographs have been taken
- ☐ All conduit labels match the Section 11 master schedule
- ☐ All conduit ends are capped with masking tape
- ☐ All draw strings are in place
- ☐ FFL chalk line is visible on every wall
- ☐ DB + starter cupboard layout choice (Option 1 / Option 2) is decided
- ☐ Walnut shelf LED driver location choice is decided OR deferred to electrician's discretion
- ☐ 6-gang vs 7-gang vs 8-gang switch panel choice is decided on-site
- ☐ Reolink back box choice is finalised (or delegated to electrician)

[X] I AUTHORISE the plaster crew to begin Phase 5.

Homeowner signature: _____

Electrician acknowledgement: _____

Date: _____

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Flagged Open Items

Items NOT covered in this plan. Some may need conduit provisioning later; others can be designed in separate sessions.

#	Item	Status	Cost when added
1	Smart door lock for 'tap to unlock'	Battery deadbolt = no conduit needed now. Wired strike/maglock = conduit needed pre-plaster.	Rs. 12,000-15,000
2	Dining hall speaker provision	Decide in next 1-2 days if you want this (before dining hall plaster)	Rs. 3,000-5,000
3	Waveshare indoor unit full hardware + dashboard	Conduit + Cat6 ALREADY provisioned in this plan. Hardware design = separate session.	Rs. 24,000
4	Door grill on East jamb (outside)	Decorative only - decide in finishing phase	varies
5	Switch plate brand	Schneider Unica White house-wide (locked 2026-05-23, Aqara H1 dropped)	Rs. 650 (foyer 18M)
6	Walnut shelf LED driver location	Option 1 (above ceiling) vs Option 2 (under shelf) - electrician decides on-site	no cost diff
7	Reolink back box exact spec	Electrician's discretion (default 3x3 GI MS modular)	Rs. 100-300
8	DB + starter cupboard final layout	Arrangement Option 1 vs 2 - on-site decision	no cost diff

Cross-references

Document	What it covers
conduits-and-cavities.md	General conduit rules + whole-house conduit schedule
water-automation-conduits.md	Water automation system (5 conduits terminate in foyer DB cupboard)
db-layout.md	DB internal circuit layout (whole house)
ground-floor-electrical.md	All other GF electrical items
materials-checklist.md	Broader house materials list
decisions/decision-log.md	Locked design decisions across the project

End of Document

Foyer Electrician Master Plan v1.1

What to do RIGHT NOW (Phase 1 is imminent)

1. PRINT Section 17 (sign-off checklist). Take it to site tomorrow morning.
2. PRINT Section 11 (conduit schedule). Give to electrician for reference during install.
3. PRINT Section 16 (build order). Share with all trades.
4. ORDER NOW: Reolink Video Doorbell PoE (3 day shipping; want it on-site by Phase 11).
5. ORDER NOW: Halo LED strip (24V 2200K 5m roll) - less common locally, order online.
6. ORDER NOW: Visaton FR 8 speaker driver.
7. BUY LOCALLY: All conduits + wires + back boxes + MCBs from Chitradurga electrical store.
8. CALL: schedule electrician for Phase 1 (marking) and Phase 2 (cutting).
9. ARRANGE: tile contractor timing - floor-route conduits MUST be laid BEFORE tiles.
10. PHOTOGRAPH: take 'before' photos of every wall before any cutting starts.

Document version 1.1 - 2026-05-22 - Ganesh Prasad, Chitradurga

If something on-site doesn't match a section, STOP and call the homeowner before proceeding.